

AUTONOMOUS ML RESEARCH / TECHJAM 2026

# SciOdyssey: Long-Horizon Autonomous ML Research Agent

Autonomous ML discovery through **pure evidence**.

PERSISTENT WORLD

FRESH SCIENTIST

EVIDENCE FIRST

Problem → Motivation → System → Experience → Evaluation → Insights

# Can an agent carry a research task to completion?

Turn a task contract into a tested model, a reproducible implementation, and evidence another researcher can inspect.

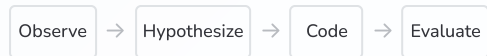
|         |                                                     |                     |
|---------|-----------------------------------------------------|---------------------|
| Explore | Choose hypotheses, write code, and run experiments. | Scientific judgment |
| Recover | Handle failures and revise unsuccessful ideas.      | Long-running work   |
| Deliver | Retain a valid model and the evidence behind it.    | Verifiable progress |

Our test case: short-video ranking on KuaiRand-Pure, with a fixed evaluator and public-validation boundary.

# Why Current Research Agents Fall Short

Three obstacles to sustained autonomous research.

## 01 CLOSED-WORLD WORKFLOW



Predefined tools and recovery paths cannot handle the unexpected.

Unexpected problems require developer intervention.

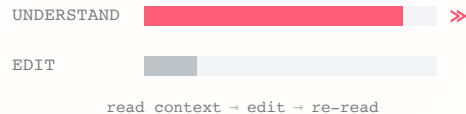
## 02 SINGLE TRAJECTORY



A long-lived context carries yesterday's momentum into today's search.

More iterations  $\neq$  more exploration.

## 03 TREE SEARCH



Reconstructing file context costs far more tokens than the edit itself.

Context cost forces repeated reading.

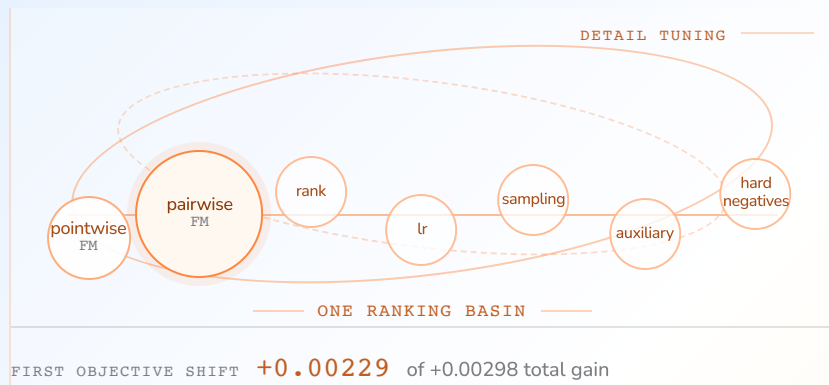
# One task. Two search styles.

Codex stays in a winning basin. Gemini keeps reopening the model.

CODEx · 11 CYCLES

0.6015 → 0.6045

## Local refinement



After the first win, later cycles stay near the same ranking basin—tuning rank, learning rate, sampling, auxiliary feedback, and negatives.

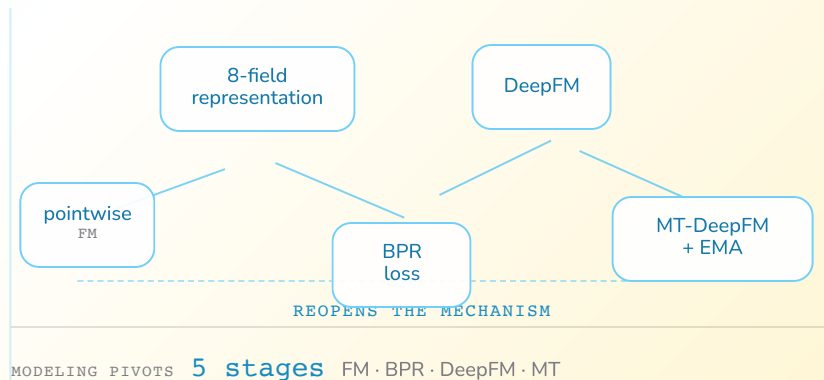
MOMENTUM IS MODEL-DEPENDENT

Codex accumulates local commitment. | Gemini keeps reopening the mechanism.

GEMINI · 8 STAGES

0.6015 → 0.6045

## Mechanism pivots



The run repeatedly changes representation, objective, or model family before a long chain of local tuning can form.

Research is an **open-world** task.  
We need an **open-world** coding agent.

### 01 OPEN ACTION SPACE



Shell



Files



Browser



Git

Use what exists

### 02 SELF-RECOVERY



Read errors



Inspect docs



Inspect  
source



Install  
packages

Fix what breaks

### 03 RUNTIME EXTENSION



MCP



Skills



CLI



Packages

Add what's missing

#### WORKFLOW EXAMPLE · LANGGRAPH

**LangGraph** makes a known workflow explicit as nodes + edges.

Missing capability? Leave the graph → inspect → extend → retry

Preserve the  
research world.

Reset the researcher.

Pass evidence, failures, and code to a fresh Scientist.

# Two scientific roles. One safe runtime.

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## Scientist

Hypothesize · code · test · interpret

One trajectory

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## META

Audit · preserve · hand off

Across trajectories

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## Runtime

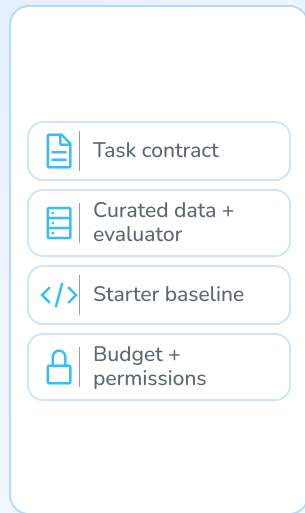
Isolation · cancellation · invariants

System lifetime

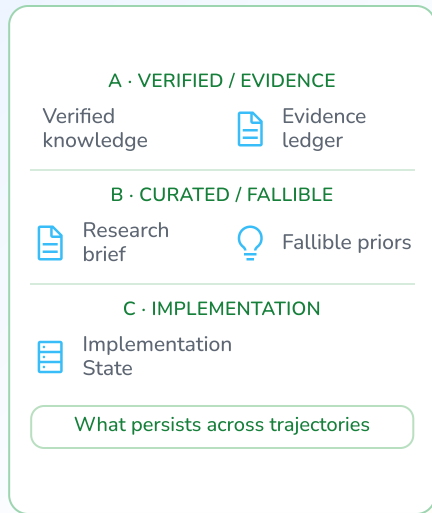
The Scientist chooses the science. META maintains the world. Runtime enforces mechanics, not scientific judgment.

# The persistence boundary is explicit.

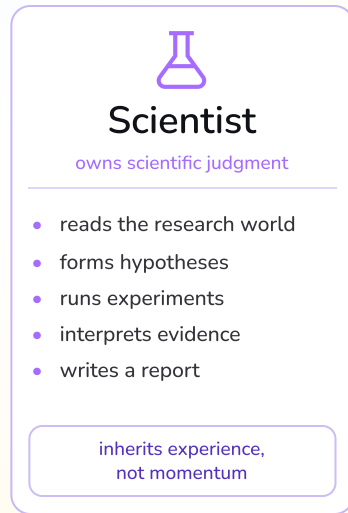
## Environment



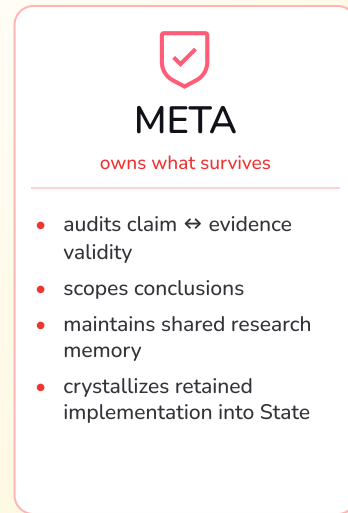
## Research World



## Fresh Scientist



## META Review



Selective persistence / trajectory handoff

audit → scope → preserve

Runtime — deterministic mechanics



Workspace isolation



Runner + evaluator



Logging



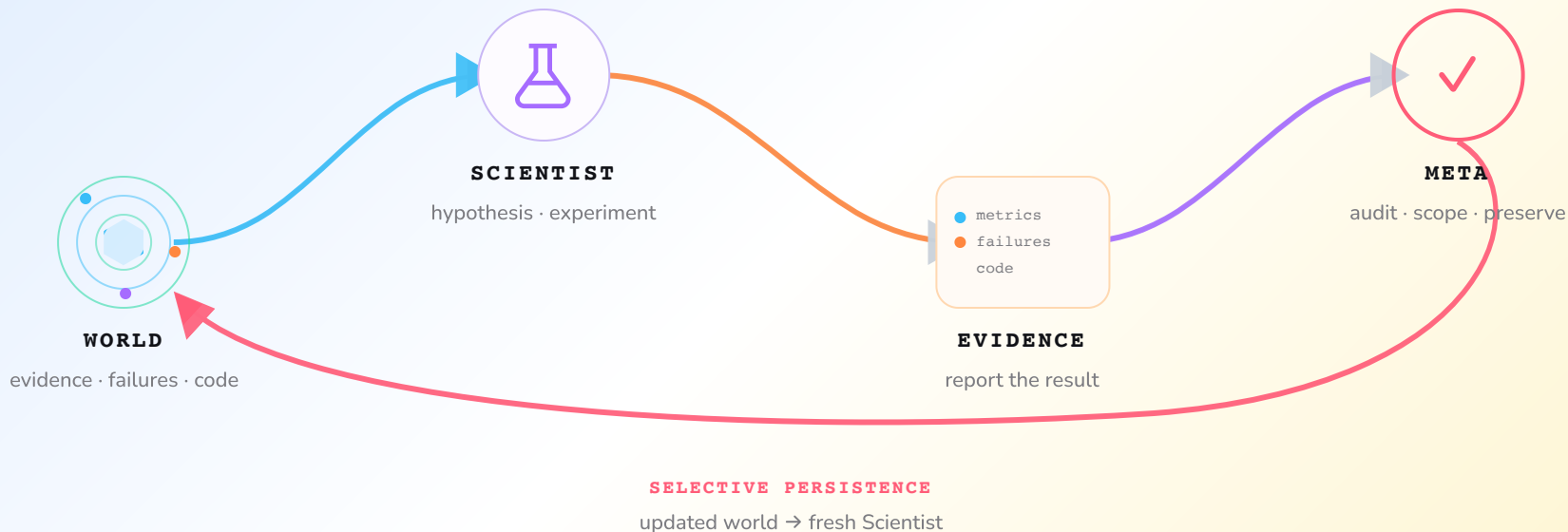
Artifact capture



State operations



# Each cycle leaves a world to build on.



**PERSISTS** metrics · code · failures · original reports

**RESTARTS** an independent reasoning trajectory

# Keep facts distinct from intuition.

research world

- └ evidence & provenance
- └ verified knowledge
- └ current research state
- └ fallible priors
- └ State + implementation

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**Built to be re-examined.**

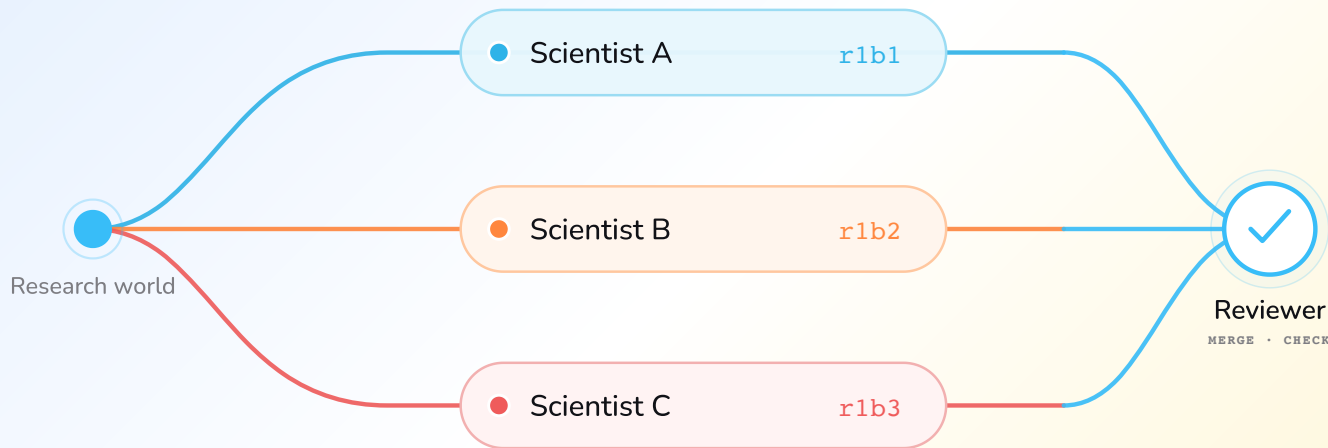
Facts have evidence.

Priors can be overturned.

Code can be picked up again.

Original reports remain available for audit.

# One starting point. Independent answers.



Explore in isolated worktrees. Review after completion.

Optional synthesis: one implementation parent + reference evidence from other branches.  
Adoption is explicit via parallel-promote. Code and memory are not merged automatically.

# Start with the task and working constraints.

The operator defines the objective and constraints; the agent chooses and executes the experiments.

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01 task.md

## What to solve.

Objectives · constraints · evaluation

### EXAMPLE

Rank short videos for each user and interaction context.

KuaiRand-Pure · long\_view · GAUC + nDCG@5

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02 PERSONAL.md

## How you want it solved.

Preferences · research style · priorities

### EXAMPLE

Run on macOS / Apple Silicon with uv.

≤ 15 min / experiment · never print credentials

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Autonomous by default. Supervisable by design.

# Evidence changed the next experiment.

E001–E013: reject weak ideas, improve representations, then test ensembles.

|      |                                                           |                  |
|------|-----------------------------------------------------------|------------------|
| E001 | Screen historical target encodings; reject weak variants. | Medium screening |
| E003 | Establish a valid rich-FM checkpoint.                     | Full · 0.6016310 |
| E008 | Retain a five-seed, 38-field FM ensemble.                 | Full · 0.6040901 |
| E013 | Select the eight-seed, 46-field FM ensemble.              | Full · 0.6059363 |

These are milestones within four autonomous cycles. Medium screens ideas; Full evaluations support retained score claims.

# An evaluation failure became a recoverable step.

Cycle 1 completed training and evaluation, then failed while writing the evidence file.

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## Failure

NumPy float32 values broke JSON serialization.

Evidence writer

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## Repair

The Scientist converted scalars and reran.

Agent-authored fix

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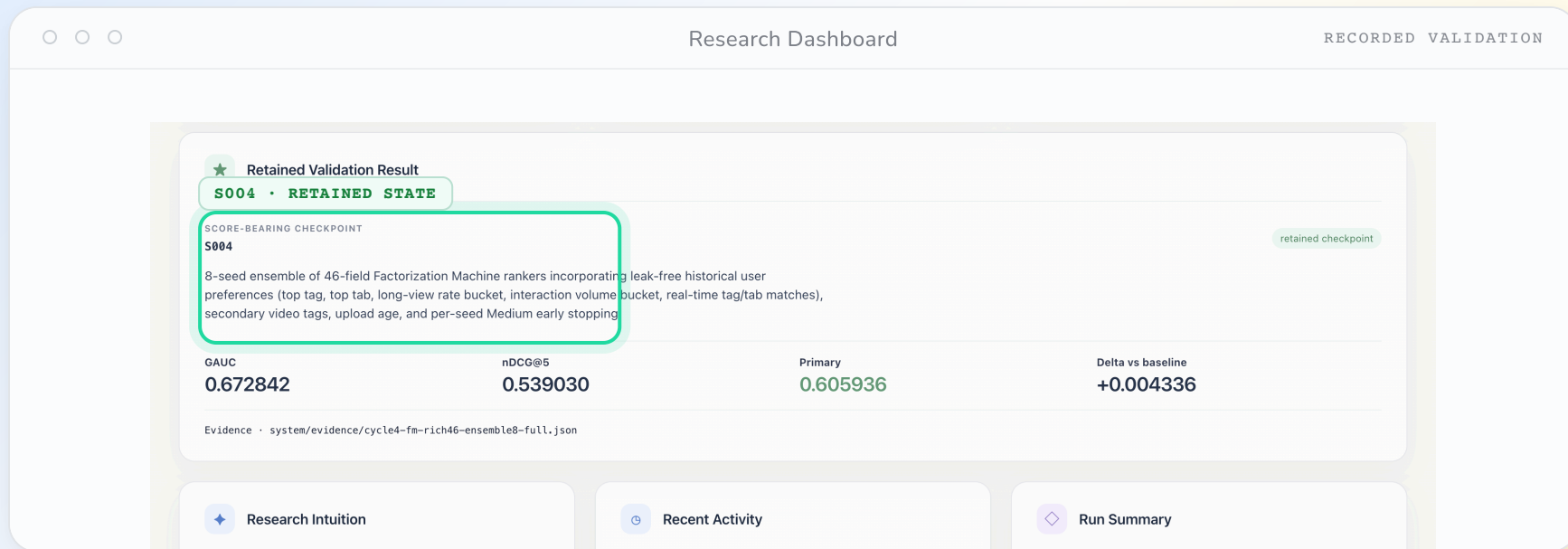
## Retain

Printed measurements and saved evidence survived.

Auditable recovery

No human selected this repair or edited the code. This is one observed recovery case, not a guarantee for every failure.

# A retained state keeps the evidence inspectable.



Inspect the retained state and the evidence behind it.

Retained validation checkpoint · evidence path · reproducible implementation.

# Autonomous research. Measured gains.

PRIMARY / RESEARCH AGENT

0.6059363

+0.0043363 absolute improvement

| METRIC  | Official FM | Ours      |
|---------|-------------|-----------|
| GAUC    | 0.6674000   | 0.6728421 |
| nDCG@5  | 0.5357000   | 0.5390304 |
| Primary | 0.6016000   | 0.6059363 |

Primary = mean(GAUC, nDCG@5). Public validation only; hidden-test scoring is organizer-controlled.

Final recipe: 46 categorical features × 8-seed FM ensemble · NumPy / CPU



# An audit trail from experiment to output.

4

Autonomous cycles

Within a 50-iteration cap

13

Named experiments

E001–E013 · 7 Full evaluations

0

Manual scientific interventions

After launch · 0 GPU-hours

170,588 prediction rows passed the unchanged Starter Kit alignment checker.

LLM tokens: 48.24M including cache reads; 4.02M excluding cache reads.

# Resource use is part of the evidence.

Project-level subscription estimate, with run-level telemetry reported separately.

~\$10

from first line of code to final result

coding · debugging · agent runs · all experiments

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## MODEL

One GPT Plus + One Gemini Pro for 7 days

Two consumer subscriptions covered the entire research loop.

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## COMPUTE

1 × MacBook M2

All development and experiments ran on a single consumer laptop.

The recorded runs used CPU training; LLM usage remains a separate cost.

# Compare scores alongside their evidence.



Runs differ in model, budget, and evidence quality; this is not a controlled causal comparison.

# The run taught us more than the final score.

What survived the experiments — and what remains unsettled.

## 01 Diversity is a prior.

Keep `gemini-3.7-flash` as META while Scientist rotates across `gpt-5.6-sol`, `gemini-3.7-flash`, and `gpt-5.6-luna`. This widens priors; it does not prove the gain.

## 03 Failure can be evidence.

After evaluation, NumPy `float32` values broke serialization. The Scientist fixed the writer, reran, and kept the measurements.

## 02 A score is not a claim.

Audits found leakage and sampling mismatches in competing experiments. META can scope evidence; it is not a formal verifier.

## 04 Reset $\neq$ no anchor.

Fresh cognition reopens the search, but shared summaries can still omit context or anchor future work. Share evidence later than cognition.

### OBSERVED, NOT PROVEN

Parallel/Synthesis reached public-validation Primary 0.6055536; the canonical E001–E013 frontier remains 0.6059363.

4 cycles

0 post-launch interventions

0 GPU-hours

# Reproducible evidence. Open questions.

|    |                                          |                    |
|----|------------------------------------------|--------------------|
| 01 | One task does not establish generality.  | External validity  |
| 02 | META, State, and reset are not isolated. | Causal attribution |
| 03 | Summaries can lose context and anchor.   | Memory quality     |

Next questions: broader tasks, targeted ablations, and less repeated execution.

# A clean repo. A continuous agent.

FIELD NOTE / 01

Engineering practices to apply next: isolate parallel changes and make unattended work reviewable.

## 01 MANY AGENTS



### Give every agent a branch.

**main**

clean trunk

● agent A / branch

● agent B / branch

● agent C / branch

review

→ merge

Let agents work independently. Conflicts stay local, and the main branch stays clean.

## 02 NO QUOTA / NO LAPTOP



### Leave the next move in GitHub.

01

Issue /  
prompt

inject the task



02

GitHub  
Action

run the agent



03

Commit

review the change

Connect the agent to GitHub Actions. It can code and commit while you are offline; review before merging.

# Keep the experience. Reopen the search.

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```
# From the configured repo, run one autonomous cycle
$ ./scripts/research-agent step --cli codex \
  --target /absolute/path/to/project --allow-edits
```

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[github.com/zc6600/research-agent-kuairand](https://github.com/zc6600/research-agent-kuairand) ↗

Code · Technical report · Experiment ledger · Checked output

# Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD



Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture ·  
integration



Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles ·  
evidence review



Shilin Xu

DATA / METRICS

Data workflow · contracts  
· alignment



GE GAO

RUNTIME / RELIABILITY

Execution · testing ·  
integration support



Jiran Li

EVIDENCE /  
REPRODUCIBILITY

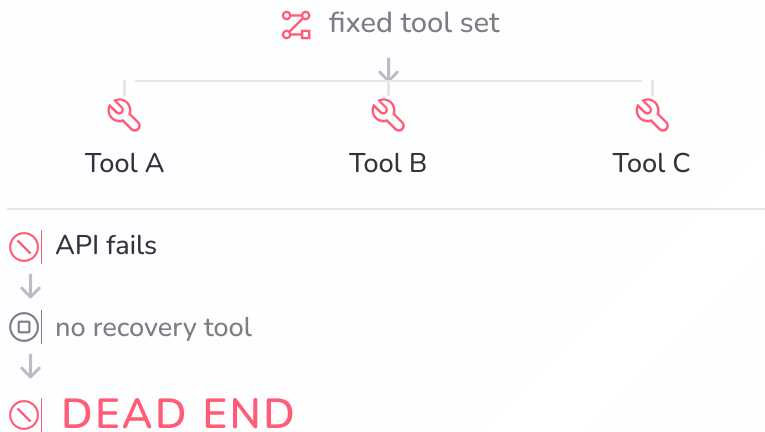
Telemetry · documentation  
· packaging



Research is an **open-world** task.  
We need an **open-world** coding agent.

## LANGGRAPH-STYLE ORCHESTRATION

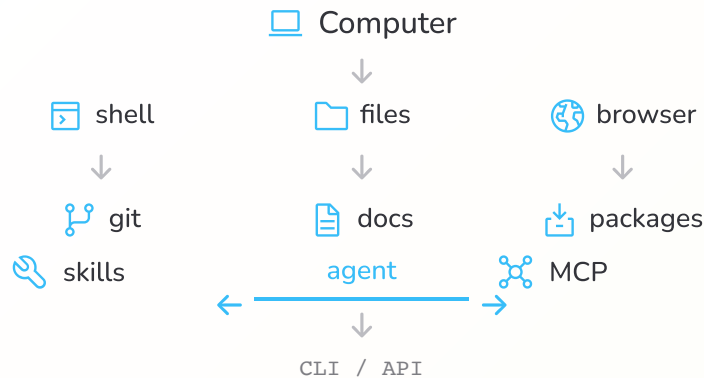
predefined graph



Capabilities are bounded by what developers anticipated.

## CODING-AGENT HARNESS

open action space



fail → inspect → learn → modify environment → retry

The agent can acquire the missing capability at runtime.